

## Lab 2 - Configuring Basic Single-Area OSPFv2

### Answer Sheet

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### Part 2: Configure and Verify OSPF Routing

#### Step 1: Verify OSPF neighbors and routing information.

b.

What command would you use to only see the OSPF routes in the routing table?  
show ip route ospf

### Part 4: Configure OSPF Passive Interfaces

- g. Re-issue the **show ip route** and **show ip ospf neighbor** commands on R1 and R3, and look for a route to the 192.168.2.0/24 network.

What interface is R3 using to route to the 192.168.2.0/24 network?

Serial0/0/1

What is the accumulated cost metric for the 192.168.2.0/24 network on R3?

65 (serial link cost 64 and Ethernet cost 1)

Does R2 show up as an OSPF neighbor on R1?

No

Does R2 show up as an OSPF neighbor on R3?

No

What does this information tell you?

Passive interfaces cause neighbor relationships to break down, and other routers cannot learn routes through this router.

- h. Change interface S0/0/1 on R2 to allow it to advertise OSPF routes. Record the commands used below.

R2(config)# router ospf 1

R2(config-router)# no passive-interface s0/0/1

- i. Re-issue the **show ip route** command on R3.

What interface is R3 using to route to the 192.168.2.0/24 network?

Serial0/0/1

What is the accumulated cost metric for the 192.168.2.0/24 network on R3 now and how is this calculated?

65 (serial link cost 64 and Ethernet cost 1)

Yes

## Part 5: Change OSPF Metrics

### Step 1: Change the reference bandwidth on the routers.

- a. To reset the reference-bandwidth back to its default value, issue the **auto-cost reference-bandwidth 100** command on all three routers.

```
R1(config)# router ospf 1
```

```
R1(config-router)# auto-cost reference-bandwidth 100
```

```
% OSPF: Reference bandwidth is changed.
```

```
Please ensure reference bandwidth is consistent across all routers.
```

Why would you want to change the OSPF default reference-bandwidth?

The default reference bandwidth is 100 Mbps, which isn't enough to properly distinguish between gigabit links. Adjusting the reference bandwidth lets the OSPF cost better reflect the real speed of each link.

### Step 2: Change the bandwidth for an interface.

g.

Explain how the costs to the 192.168.3.0/24 and 192.168.23.0/30 networks from R1 were calculated.

192.168.3.0/24: cost of R1–R3 serial link (781, because bandwidth 128 kbps) + R3 G0/0 cost (1) = 782    192.168.23.0/30: cost of R1–R3 serial link (781) + R3–R2 serial link (64 default bandwidth on R3 side) = 845

- i. Issue the **bandwidth 128** command on all remaining serial interfaces in the topology.

What is the new accumulated cost to the 192.168.23.0/24 network on R1? Why?

1562

(All serial links now have bandwidth 128 kbps → cost 781 each; path cost = 781 + 781)

### Step 3: Change the route cost.

c.

Explain why the route to the 192.168.3.0/24 network on R1 is now going through R2?

The ip ospf cost 1565 command on R1's S0/0/1 makes the direct path to R3 more expensive (1565 + 1 = 1566) than the path through R2 (781 + 781 + 1 = 1563). OSPF therefore installs the lower-cost route via R2.

### Reflection

1. Why is it important to control the router ID assignment when using the OSPF protocol?

The router ID uniquely identifies the router within the OSPF domain. If it changes—for example, due to an interface IP address change—adjacencies will reset and the SPF algorithm will run again, which causes instability. Manually setting the router ID (using a loopback interface or the router-id command) keeps it stable.

2. Why is the DR/BDR election process not a concern in this lab?

All inter-router links are serial point-to-point (no DR/BDR election is needed). The LAN interfaces (G0/0) only connect to PCs, not to other OSPF routers, and will later be configured as passive interfaces.

3. Why would you want to set an OSPF interface to passive?

It prevents hello packets and routing updates from being sent on the LANs that have no other OSPF routers. This saves bandwidth and CPU resources, reduces unnecessary traffic, and improves security by keeping routing protocol packets off the LAN.

4. What aspect of OSPF makes the protocol hierarchical and permits the creation of very scalable networks?

The use of areas in multi-area OSPF.

5. What single OSPF router configuration command allows the assignment of OSPF area 0 to all interfaces in the range 10.0.0.0 to 10.255.255.255?

```
network 10.0.0.0 0.255.255.255 area 0
```

6. A router has three routes to the same network: one route from a RIP with a metric of 4, another from OSPF with a metric of 3053092, and another from EIGRP with a metric of 4039043. Which of the three routes will be used for the routing decision?

The EIGRP route is preferred because it has the lowest administrative distance (90), compared to OSPF (110) and RIP (120). Administrative distance is always checked before the metric.